Armin Iranpour



Mechanical Engineer | M.Sc. TU Dortmund | Production & Logistics Planning · Mechanical Design (CATIA V5) · Material Flow Simulation

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SUMMARY

Master's student in Mechanical Engineering at TU Dortmund (IT in Production and Logistics, GPA 1.4 on the German scale) with about 4 years of experience designing multi-disc clutches for the oil and gas industry, working closely with production, purchasing and assembly. Focus on material flow simulation, supply chain management and data analysis. Looking for a working student role or an entry-level position in production, design or logistics in the Dortmund area.

EXPERIENCE

09/2020 – 08/2024

Product Designer / Design Engineer

Faraz Foolad Sanati Espadana Co. · Full-time · Isfahan, Iran

Design of multi-disc clutches for oil and gas plants, at the interface to production, purchasing and assembly.

- Designed 25+ assemblies for manufacturability in CATIA V5, incl. manufacturing drawings and BOMs
- Wrote routing sheets and work instructions; cut assembly time per unit by about 15%
- Derived material requirements for around 40 orders per year; reduced ordering errors by about 20%
- Coordinated delivery dates with 15 suppliers; raised on-time delivery to about 90%
- Tracked lead times and removed bottlenecks with plant management; cut lead time by about 10%
- Analysed field failures with 5-Why and Ishikawa; extended disc service life by about 20%
- Supported on-site assembly and commissioning

Skills: CATIA V5 · Production planning · Process planning · Microsoft Excel · Root-cause analysis

06/2019 – 09/2019

Intern, Manufacturing and Logistics

Esfahan Machinery Manufacturing Co. · Internship · Isfahan, Iran

- Mapped material and information flows between warehouse and production; documented waiting times at 6 workstations
- Prepared and checked technical drawings for manufacturing projects

EDUCATION

10/2024 – 03/2027 (expected)

Master of Science, Mechanical Engineering

TU Dortmund University · GPA 1.4 (German scale)

IT in Production and Logistics. Focus: material flow simulation, supply chain management, production systems, data analysis, industrial information systems.

09/2016 – 09/2020

Bachelor of Science, Mechanical Engineering

University of Kashan, Iran

Thesis: technical and economic feasibility of an additively manufactured plate heat exchanger, including a cost comparison with conventional manufacturing.

PROJECTS

Enterprise Dynamics

Material Flow Simulation & Digital Twin Design Lab

TU Dortmund

Built a production-line model and evaluated 4 layout and control scenarios by throughput, utilisation and lead time; identified the bottleneck station. Result: about 12% higher simulated throughput (recommended scenario).

Python (Pandas), Excel

Data Product – Information Systems Case Study

TU Dortmund

Cleaned and analysed public logistics data (Eurostat) and built a KPI dashboard in Excel for logistics planners. Result: Dashboard presented at the chair.

Abaqus

FEM Analysis of Machine Elements

TU Dortmund

Simulated stress concentrations in a stepped shaft, validated against the analytical solution (deviation below 5%) and optimised the notch geometry. Result: Peak stress lowered by about 18%.

SKILLS

Mechanical design	Mechanical design, CATIA V5, technical drawing (ISO), finite element analysis (Abaqus, ANSYS), machine elements, strength calculation, drive technology, material selection, design for manufacturing, manufacturing technology, assembly, commissioning
Planning & production	Production planning, process planning, logistics planning, supply chain management, lean manufacturing, process optimisation
Simulation & data	Material flow simulation (Enterprise Dynamics), MATLAB, Microsoft Excel, Python, SQL
Methods	Root-cause analysis (5-Why, Ishikawa), reliability engineering, cross-functional collaboration

LANGUAGES

German	Professional working proficiency (C1, TestDaF TDN 5)
English	Fluent
Persian	Native